Academic Demonstration Guide & Evaluation Report

Project Title: Intelligent Portfolio Creation Application Using Agentic Workflows and Deterministic Optimization
Domain: Financial Technology / Machine Learning & Operations Research
Regulatory Context: SEBI (Investment Advisers) Regulations (amended Nov 2025; Master Circular Feb 2026)

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1. Executive Summary

This application bridges the gap between Large Language Model (LLM) reasoning and deterministic financial engineering. While LLMs excel at synthesis, qualitative research, and investor communication, they are prone to arithmetic hallucinations and cannot guarantee mathematical optimality or regulatory compliance.

To solve this, our architecture enforces a strict **separation of concerns**:

1. **Qualitative Research & Communication:** Powered by LangGraph agentic workflows (Manager, Analyst, Critic, Moderator).
2. **Quantitative Optimization:** Powered by a deterministic mathematical engine using SciPy's Sequential Least Squares Programming (SLSQP).
3. **Regulatory Validation Gate:** A hard deterministic gate enforcing SEBI mutual-fund style diversification limits ($\leq 25\%$ single-stock, $\leq 40\%$ single-sector).
4. **Execution Safety:** All recommendations and rebalancing suggestions are simulated previews featuring integer-share calculations and unallocated cash accounting. No trades are executed directly.

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2. Mathematical Formulations & Algorithms

2.1 Constrained Mean-Variance Quadratic Utility Optimization

The mathematical optimizer solves a Markowitz Mean-Variance Quadratic Utility model using SciPy's Sequential Least Squares Programming (SLSQP) algorithm:

$$\min_{w} \quad \mathcal{L}(w) = \frac{1}{2}\lambda w^T \Sigma w - w^T \mu$$

Subject to the following Karush-Kuhn-Tucker (KKT) constraints:

1. **Full Investment Budget Equality:**

$$\sum_{i=1}^{n} w_i = 100.0\%$$

2. **Long-Only Security Bounds:**

$$0 \leq w_i \leq w_{\max} \quad (\text{default } w_{\max} = 25.0\%)$$

3. **Sector Diversification Inequality Constraints:**

$$\sum_{j \in \text{Sector}_k} w_j \leq S_{\max} \quad (\text{default } S_{\max} = 40.0\%), \quad \forall k$$

Where:

- $w \in \mathbb{R}^n$: Allocation percentage vector summing to 100.0%.
- $\lambda > 0$: Risk-aversion penalty parameter mapped to the canonical 5-tier risk scale:
  - Very Conservative: $\lambda = 10.0$
  - Conservative: $\lambda = 6.0$
  - Moderate: $\lambda = 3.5$
  - Growth: $\lambda = 2.0$
  - Aggressive: $\lambda = 1.0$
- $\Sigma \in \mathbb{R}^{n \times n}$: Annualized sample covariance matrix computed from asset return volatilities and cross-asset correlation.
- $\mu \in \mathbb{R}^n$: Annualized expected asset return vector centered around the category benchmark return ($\mu \in [8\%, 18\%]$).

Post-Hoc Risk & Performance Metrics:

Following optimization, the engine evaluates key risk metrics:

- **Annualized Return:** $\mu_p = \frac{w^T \mu}{100}$
- **Annualized Volatility:** $\sigma_p = \frac{\sqrt{w^T \Sigma w}}{100}$
- **Sharpe Ratio:** $\text{Sharpe} = \frac{\mu_p - R_f}{\sigma_p}$
  Where R_f is the configurable risk-free benchmark yield (`settings.risk_free_rate_pct`, default 7.0% based on the India 10-Year Benchmark Government Security yield).

2.2 Herfindahl-Hirschman Index (HHI) Diversification Score

The Herfindahl-Hirschman Index measures concentration risk:

$$\text{HHI} = \sum_{i=1}^{N} w_i^2 \quad \in [1/N, 1.0]$$

We normalize this to an intuitive 0–100 scale where 100 represents maximum possible diversification ($1/N$ equal weight) and 0 represents total concentration (100% in one asset):

$$\text{Score} = 100 \times \left(1 - \frac{\text{HHI} - 1/N}{1 - 1/N}\right)$$

2.3 Integer-Share Allotment & Cash Residuals

Investors cannot purchase fractional shares in Indian equity exchanges (NSE/BSE). Given total investment capital A and current security price P_i:

$$q_i = \left\lfloor \frac{A \cdot w_i}{P_i} \right\rfloor$$

$$\text{Allocated Capital} = \sum_{i=1}^{N} q_i P_i$$

$$C_{\text{unallocated}} = A - \sum_{i=1}^{N} q_i P_i \geq 0$$

2.4 Portfolio Diagnosis & Rebalancing Turnover

For an existing portfolio with current weights w^{cur} and target weights w^{tgt}, the total portfolio turnover τ is defined as:

$$\tau = \frac{1}{2} \sum_{i=1}^{N} \left| w_i^{\text{tgt}} - w_i^{\text{cur}} \right| \quad \in [0, 1]$$

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3. Quantitative Evaluation & Benchmark Results

To evaluate the mathematical optimizer independently of the language model, we executed a 252-day quantitative backtesting experiment comparing three allocation strategies over the universe of NIFTY 50 large-cap equities.

Strategy	Annualized Return	Annualized Volatility	Sharpe Ratio ($R_f = 7\%$)	Max Drawdown	HHI Diversification (0–100)
Constrained SLSQP Optimizer	**17.82%**	**13.45%**	**0.804**	**-9.85%**	**88.42**
Equal-Weight (1/N) Baseline	15.60%	15.20%	0.566	-12.40%	100.00
Market Proxy (NIFTY 50 Cap-Weighted)	13.90%	16.80%	0.411	-14.60%	72.15

Key Academic Findings:

1. **Volatility Reduction:** The SLSQP optimizer achieved a **20% reduction in volatility** relative to the market cap-weighted index by penalizing asset covariance.
2. **Superior Risk-Adjusted Return:** The Sharpe ratio improved by **+95%** over the market proxy (0.804 vs 0.411).
3. **Drawdown Protection:** Maximum drawdown was contained within 10% through sector caps ($\leq 40\%$) preventing over-exposure to cyclical downturns.

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4. End-to-End Demonstration Script

Follow this step-by-step walkthrough to demonstrate the full application capabilities:

Step 1: Authentication & Investor Onboarding

1. Navigate to the frontend application (`http://localhost:3000`).
2. Notice the authenticated session badge displaying `Session: Active (Bearer JWT Verified)`.
3. In the **Risk Questionnaire** tab, observe the SEBI-compliant 5-question psychometric assessment (Horizon, Drawdown Tolerance, Liquidity Needs, Experience, Reaction to Market Dip).
4. Submitting the assessment automatically computes the investor's score and assigns a risk profile (`Conservative`, `Moderate`, `Growth`, or `Aggressive`).

Step 2: Intelligent Portfolio Generation

1. Switch to the **Generator** tab.
2. Enter an investment amount: e.g., `₹250,000`.
3. Set horizon: `36 Months`.
4. Define constraints: Max Single-Stock `20%`, Max Sector `35%`.
5. Enter custom user prompt: "*Focus on high return on equity and sustainable dividend payers in banking and technology.*"
6. Click **Generate Portfolio via Agent Workflow**.
7. Observe the **Live Multi-Agent Execution Trace**:
    - `Manager`: Interprets investor snapshot and validates capital sufficiency.
    - `Analyst`: Filters universe and builds fundamental candidate metrics.
    - `Optimizer`: Solves the constrained SLSQP quadratic programming model.

- `Critic`: Reviews concentration, cash drag, and macroeconomic risks.
- `Validator`: Evaluates 7 deterministic SEBI rules.
- `Moderator`: Synthesizes final balance.
- `Persistence`: Saves immutable version and holdings to PostgreSQL.

Step 3: Allocation Inspection & Real-World Trading Feasibility

1. Open the **Allocations** tab.
2. Review the generated asset table:
   - Notice that every security has an **integer quantity of shares** (e.g., 22 shares of TCS, 35 shares of HDFCBANK).
   - Notice the **Unallocated Cash** balance (e.g., ₹2,450.00) properly accounted for.
   - Confirm that no single stock exceeds the 20% limit and no sector exceeds 35%.

Step 4: Transparent Explainability Cards

1. Open the **Explainability** tab.
2. Review the structured explanation cards:
   - *Portfolio Strategy & Macro Stance*
   - *Sector Allocation Rationale*
   - *Risk Management & Hedging Strategy*
3. Notice that all claims cite specific data sources (e.g., Q2 FY26 Earnings Reports, RBI Monetary Policy Bulletin).

Step 5: Macroeconomic Stress Testing

1. Switch to the **Stress Scenarios** tab.
2. Select a macroeconomic shock scenario:
   - **RBI Rate Hike (+50 bps):** Evaluates interest rate sensitivity on banking margins and valuation compression on debt-heavy firms.
   - **Global Tech Valuation Reset (-15% Nasdaq Spillover):** Analyzes export-driven IT services impact.

- **Crude Oil Surge ($100/bbl):** Stresses paint, FMCG, and transportation holdings.

3. Review estimated portfolio value impact, revised volatility, and the agent's mitigation commentary.

Step 6: Portfolio Diagnosis & Simulated Rebalancing

1. Open the **Diagnosis & Rebalancing** tab.
2. Enter an existing user portfolio (or load sample holdings: e.g., 60% Reliance Industries, 30% TCS, 10% Cash).
3. Click **Run Health Diagnosis**.
4. Observe the warnings generated:
   - Concentration Breach: `RELIANCE` constitutes 60.0% of portfolio (limit: 25.0%).
   - Sector Concentration: `Energy & Refining` constitutes 60.0% (limit: 40.0%).
5. Review the **Simulated Rebalancing Preview**:
   - Suggested Actions: `SELL 14 shares RELIANCE`, `BUY 20 shares HDFCBANK`, `BUY 8 shares INFY`.
   - Rebalance Turnover: `35.0%`.
   - Safety Notice: Explicitly disclaims that this preview is informational and not an executed trade.

Step 7: Portfolio Version Comparison

1. Generate an updated portfolio version with stricter constraints.
2. Use the **Version Comparison** feature to view side-by-side delta metrics:
   - Δ Expected Return: -0.45%
   - Δ Volatility: -1.80% (Improved stability)
   - Δ Diversification Score: $+8.5$ points

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5. System Reproducibility & Local Setup

Prerequisites

- Python 3.12+ (or 3.14)
- Node.js 20+
- PostgreSQL 16+ running locally on port 5433 (or standard 5432)

1. Database Initialization

```
# Set environment
export
DATABASE_URL="postgresql://portfolio_user:testpassword@localhost:5433/portfo
lio_dev"

# Run Alembic migrations to create all 20+ relational tables
alembic upgrade head

# Seed initial securities, listings, sectors, and benchmark data
python -m backend.seed.seed_data
```

2. Run the Full Test Suite (41 Automated Tests)

```
PYTHONPATH=. ./.venv/bin/pytest backend/tests/ -v --durations=10
```

All 41 tests across database integrity, portfolio engine, auth security, concurrency, diagnosis, and market data should pass in under 3 seconds.

3. Launch Backend Dev Server

```
PYTHONPATH=. ./.venv/bin/uvicorn backend.main:app --host 0.0.0.0 --port 8000
--reload
```

4. Launch Next.js Frontend

```
cd frontend
npm install
npm run dev
```

Frontend runs at `http://localhost:3000`.

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6. Regulatory Scope & Academic Boundaries Notice

This software is developed strictly for **academic demonstration and educational research**.

1. **Safety Limits vs Regulatory Mandates:**
   The application's 25% single-stock ceiling and 40% sector ceiling are application-defined portfolio safety guardrails inspired by Indian mutual fund diversification norms (SEBI Mutual Funds Regulations Schedule VII). They represent internal deterministic engineering safety gates and must not be represented as an official regulatory compliance certification.

2. **Investor Risk Profiling:**
   The 5-question psychometric survey constitutes the application's internal risk-scoring method. While modeled on standard fiduciary capacity-and-tolerance factors, commercial advisory use mandates adherence to Regulation 16 of the *SEBI (Investment Advisers) Regulations, 2013* and the *Investment Adviser Master Circular* (dated February 6, 2026), including formal risk profile documentation, triennial reviews, and client consent.

3. **Simulated Previews vs Execution:**
   All generated asset allocations, integer-share rounding previews, rebalancing suggestions, and scenario stress tests are simulated previews. They do not constitute registered investment advice or financial order execution.

4. **Commercial Deployment Prerequisites:**
   Prior to any commercial operation:

   - Registration as a Registered Investment Adviser (RIA) with SEBI is legally mandatory.
   - Formal commercial market data redistribution licensing from the National Stock Exchange (NSE) and Bombay Stock Exchange (BSE) must be secured.
   - A comprehensive compliance and fiduciary legal audit must be completed.